28/01/25

Q- A Class had a strength of 40 students numbered from 1 to 40. Students with even roll numbers were doing English literature, while students with roll numbers in multiples of 4 were doing Fashion Design, and students with roll numbers in multiples of 8 were in the technical English. Calculate the number of students who opted for none of the three streams?

Sol=

$S = \{1, 2, 3, 4 \ldots\ldots 40\}$

$n(S) = 40$

$n(E) = \frac{\cancel{40}^{20}}{\cancel{2}} = 20$

$E = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20 \ldots\ldots 40\}$

$n(F) = \frac{\cancel{40}^{10}}{\cancel{4}} = 10$

$F = \{4, 8, 12, 16, 20, 24, 28, 32, 36, 40\}$

$n(T) = \frac{\cancel{40}^{5}}{\cancel{8}} = 5$

$T = \{8, 16, 24, 32, 40\}$

$(E \cap F) = \{4, 8, 12, 16, 20, 24, 28, 32, 36, 40\}$

$n(E \cap F) = 10$

$F \cap T = \{8, 16, 24, 32, 40\}$

$n(F \cap T) = 5$

$E \cap T = \{8, 16, 24, 32, 40\}$

$n(E \cap T) = 5$

$E \cap T \cap F = \{8, 16, 24, 32, 40\}$

$n(E \cap T \cap F) = 5$

$$n(E \cup F \cup T) = n(E) + n(F) + n(T) - n(E \cap F) - n(F \cap T) - n(E \cap T) + n(E \cap F \cap T)$$

$$= 20 + 10 + 5 - 10 - 5 - 5 + 5$$

$$= 40 - 20$$

$$[n(E \cup F \cup T) = 20]$$

who opted for none of the three streams

$$= n(S) - n(E \cup F \cup T)$$

$$= 40 - 20$$

$$= 20$$

Q = A class had 120 students, out of which 54 students liked to have pizza and 84 students preferred to have a Salad. However Each student preferred to have at least one of the two dishes. Calculate the no of students who like both Pizza and Salad.

Sol =

$n(T) = 120$

$n(P) = 54$

$n(S) = 84$

$n(T)$ or $n(P \cup S) = n(P) + n(S) - n(P \cap S)$

$120 = 54 + 84 - n(P \cap S)$

$n(P \cap S) = 138 - 120$

$n(P \cap S) = 18$ //

No of Students who like both Pizza & Salad = 18 //

Q = In a recent survey conducted in a school it was discovered that 80 students preferred an I-Phone whereas 60 students went for Android Phone, However, 20 students were not that particular and accordingly could go with any of them, If each student preferred to have at least one of the items Find out how many students, the survey was

Sol =

$n(I) = 80$ [like I-Phone]

$n(A) = 60$ [like Android]

$n(I \cap A) = 20$ [Could go with any of them]

$$n(I \cup A) = n(I) + n(A) - n(I \cap A)$$

$$= 80 + 60 - 20$$

$$= 140 - 20$$

$$\left[n(I \cup A) = 120 \right]$$

No of Student preferred to have at least one of the items = 120

* Questions For Practice

Q-1 = There are 210 members in a club. 100 of them drink tea and 65 drink tea but not coffee. each member drinks tea or coffee. Find how many drinks coffee. How many drink coffee, but not tea.

Q-2 = In a group of 50 people, 28 have traveled to Europe, 31 have traveled to Asia, and 10 have traveled to both continents. How many people have not traveled to either continent.

Q-3 In a college of 350 students, every student need to choose among the three subjects (i.e Economics, Accounts, & Taxation) offered along with the main course. The student who chose each of these subjects are 120, 80, 95. The no of students who chose more than one of the three is 28 more than the no of students who chose all the three subjects, of three are no students who chose none of the three subjects, how many students study all three subjects?

Q-4 In a group of 300 students, 216 students study Hindi and 129 study Punjabi. How many study Hindi only? How many study Punjabi only and How many study both Hindi & Punjabi.

Q5 In a group of 120 people, 54 like Coca Cola and 84 like Pepsi and each person likes at least one of the two beverages How many like both Coca Cola & Pepsi.